

Audit Management System – Extended Use Case

(With Approval Workflow & Microservices Readiness)

1. Objective (Revised)

The Audit Management System enables organizations to **plan, execute, review, approve, and close internal audits** in a controlled, traceable, and compliant manner. The system ensures:

- Structured audit execution
 - Accountability for corrective actions
 - Multi-level approvals to maintain governance
 - Scalability using a microservices-based architecture
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2. Actors & Roles

Role	Description
Admin	System configuration, master data, audit approver
Auditor	Conducts audits, submits observations
Employee	Owns corrective actions
Audit Manager	Reviews and approves audit results
Department Head	Approves corrective action closure
System	Automated status transitions & notifications

3. High-Level Audit Lifecycle (With Approvals)

Draft Audit



Audit Scheduled (Admin Approval)



Audit In Progress



Observations Submitted



Audit Review & Approval (Audit Manager)



Corrective Actions Assigned



Corrective Action Closure Request



Corrective Action Approval (Dept Head)



Audit Closed

4. Detailed Use Cases

4.1 Audit Creation & Scheduling

Use Case: Create & Approve Audit

Primary Actor: Admin

Secondary Actor: Audit Manager

Flow:

1. Admin creates an audit in **Draft** status.
2. Audit details include:
 - Audit Name
 - Department
 - Audit Type (Internal / External / Compliance)
 - Assigned Auditor(s)
 - Planned Start & End Dates
3. Admin submits audit for approval.
4. Audit Manager reviews:
 - Scope
 - Timeline
 - Auditor assignment
5. Audit Manager:
 -  Approves → Audit status becomes **Scheduled**
 -  Rejects → Audit returns to **Draft with comments**

Approval Value: Prevents unauthorized or poorly scoped audits.

4.2 Audit Execution & Observation Capture

Use Case: Perform Audit & Log Observations

Primary Actor: Auditor

Flow:

1. Auditor accesses audits in **Scheduled** status.
2. When audit starts, system moves status to **In Progress**.
3. For each audit, Auditor logs observations:
 - Observation Title
 - Detailed Description
 - Severity (Low / Medium / High / Critical)
 - Regulation / Policy reference (optional)
 - Evidence upload (via File Service)
4. Observations are saved in **Draft** until submission.
5. Auditor submits all observations together.

System Action: Once submitted, observations become **Read-Only**.

4.3 Audit Review & Approval

Use Case: Approve Audit Findings

Primary Actor: Audit Manager

Flow:

1. Audit Manager receives notification that audit findings are submitted.
2. Reviews:
 - Completeness of observations
 - Severity classification
 - Supporting evidence
3. Decision:
 -  **Approved** → Observations are finalized
 -  **Rejected** → Sent back to Auditor for correction
4. Once approved:
 - Audit status changes to **Findings Approved**
 - Corrective Action creation is enabled

Approval Value: Ensures audit quality and reduces disputes.

4.4 Corrective Action Assignment & Tracking

Use Case: Create Corrective Actions

Primary Actor: Auditor / Audit Manager

Flow:

1. For each approved observation:
 - Corrective action is defined
 - Assigned to an Employee
 - Due Date is set
 2. Action status starts as **Open**.
 3. Employee works on the action and updates status:
 - Open → In Progress → Closure Requested
 4. Evidence of correction can be attached.
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4.5 Corrective Action Closure Approval (Second Approval Layer)

Use Case: Approve Corrective Action Closure

Primary Actor: Department Head / Audit Manager

Flow:

1. Employee submits action for closure.
2. Approver reviews:
 - Action description
 - Closure evidence
 - Impact resolution
3. Decision:
 -  **Approved** → Action status becomes **Closed**
 -  **Rejected** → Action reopens with comments
4. System checks:
 - If all actions are closed → Audit eligible for closure

Approval Value: Prevents premature or invalid action closure.

4.6 Final Audit Closure Approval

Use Case: Close Audit

Primary Actor: Audit Manager

Flow:

1. System validates:
 - All observations approved
 - All corrective actions closed
2. Audit Manager reviews final status.
3. Approves audit closure.
4. Audit status becomes **Completed & Locked**.

System Restriction: Completed audits cannot be edited.

5. Microservices Mapping (With Approvals)

Microservice	Key Responsibilities
Audit Service	Audit lifecycle, audit approvals
Observation Service	Findings, severity, submission
Approval Service	Generic approval workflows
Action Service	Corrective actions & closure
User Service	Roles & hierarchy
File Service	Evidence management
Notification Service	Approval alerts
Reporting Service	Read-only dashboards

6. Approval Service (Generic & Reusable)

Approval Types Supported

- Audit Creation Approval
- Audit Findings Approval
- Corrective Action Closure Approval
- Final Audit Closure Approval

Approval States

Pending → Approved → Rejected

Benefits:

- Same workflow engine reusable across modules
 - Role-based approvals
 - SLA & escalation possible later
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7. Reporting & Audit Trail

Audit Trail Captured:

- Who created the audit
- Who approved/rejected and when
- Observation revisions
- Action closure evidence
- Final audit sign-off

Reports Include:

- Audits pending approval
- Overdue corrective actions
- High-risk audits
- Department-wise compliance trends